CSE-5306 PROJECT 2

# Distributed Pharmacy Inventory Management System

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*gRPC Microservice Architecture vs REST Monolith*

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# Agenda


**01** **System Requirements**
6 functional + non-functional requirements

**02** **Architecture Design**
gRPC Microservice (6 nodes) + REST Monolith

**03** **Communication Model**
gRPC with Protobuf vs HTTP/JSON

**04** **Evaluation Methodology**
Benchmark setup and workload specification

**05** **Performance Results**
Latency & throughput comparison

**06** **AI Tools & Lessons**
How Claude accelerated development

# System Requirements

### 💊 Add Drug

Register new drug with name, quantity, price, expiry

### 🔍 Get Drug

Query drug details by ID

### 📦 Update Stock

Modify inventory quantity

### 🗑️ Delete Drug

Remove drug record from system

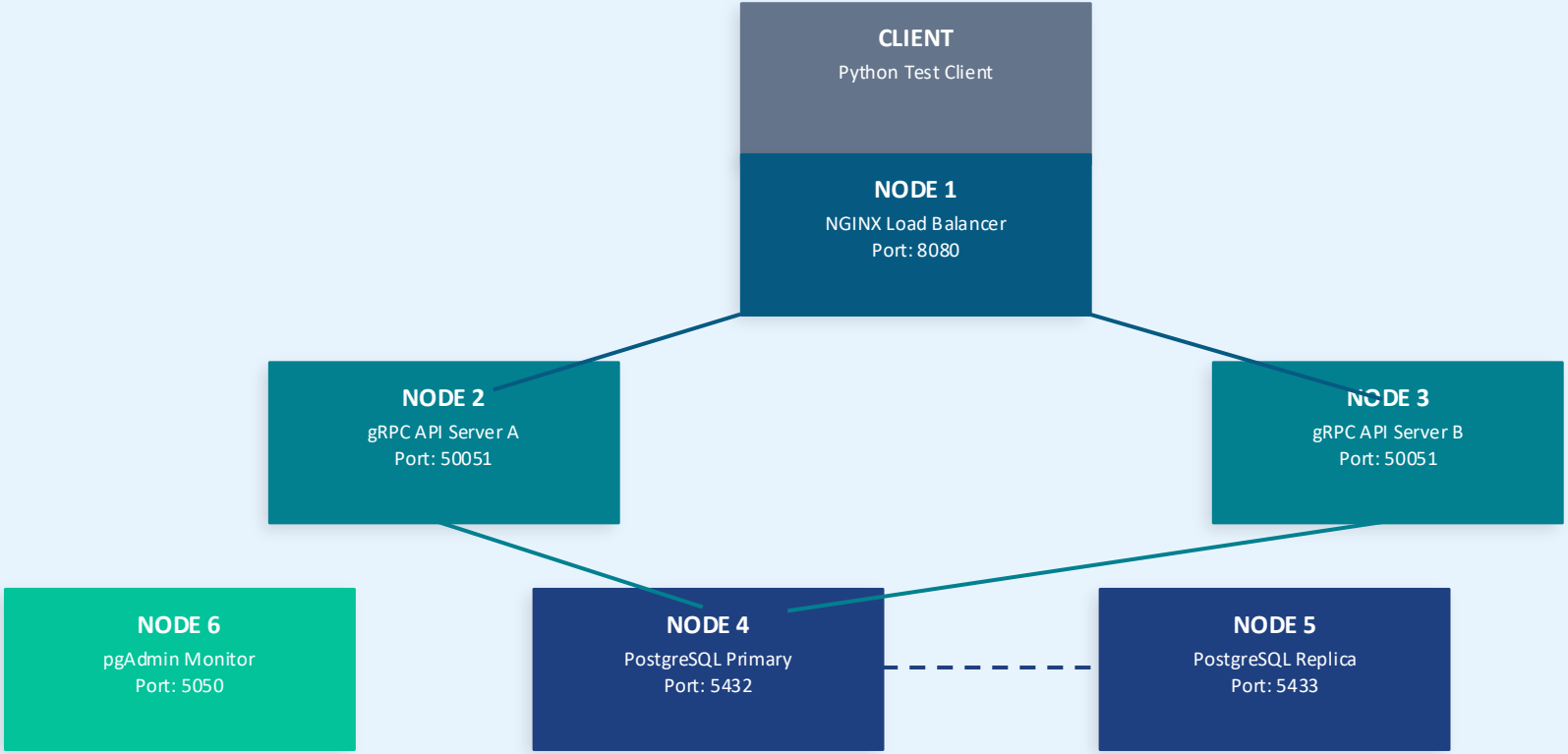
### 📋 List All Drugs

View complete inventory list

### ⚠️ Low Stock Alert

Find drugs below threshold quantity

# Architecture Design — gRPC Microservice (6 Nodes)




*--- Replication | — gRPC calls*

# Communication Model

## gRPC Microservice

**Protocol:** HTTP/2 + Protobuf binary

**Format:** Binary serialization

**Speed:** Faster encoding/decoding

**Streaming:** Bidirectional streams

**Contract:** Strongly typed .proto file

**Best for:** Internal service-to-service calls

## REST Monolith

**Protocol:** HTTP/1.1 + JSON text

**Format:** Human-readable JSON

**Speed:** Slower JSON parsing

**Streaming:** Not supported

**Contract:** Flexible, no schema required

**Best for:** Public APIs, browser clients

# Evaluation Methodology

## Hardware Environment

Machine: MacBook Pro (Apple Silicon)

RAM: 16 GB | OS: macOS

Docker Desktop (all containers local)

Python 3.9 benchmark scripts

## Workload Specification

Write: concurrent AddDrug requests

Read: concurrent ListDrugs requests

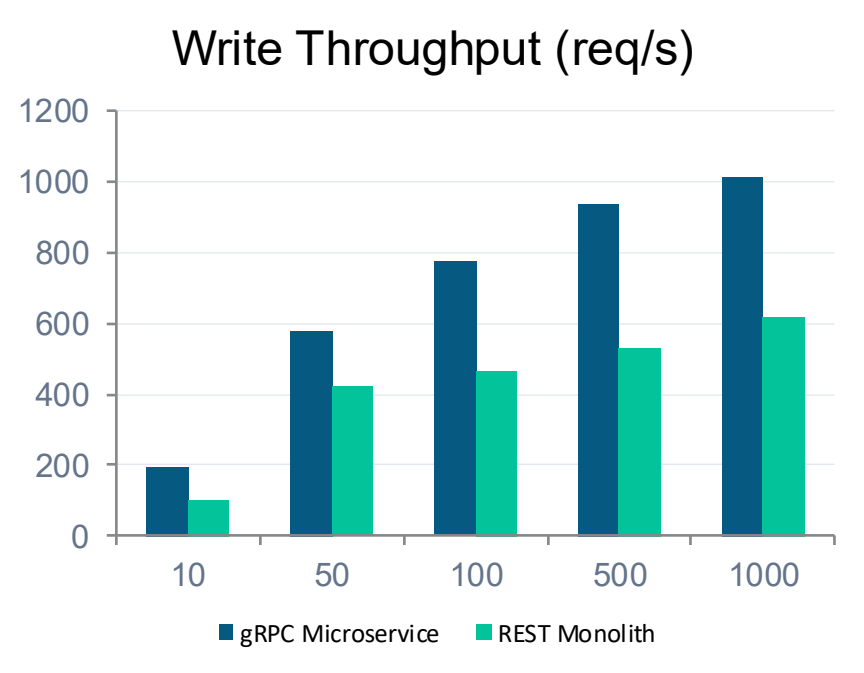
Concurrency: 10, 50, 100, 500, 1000

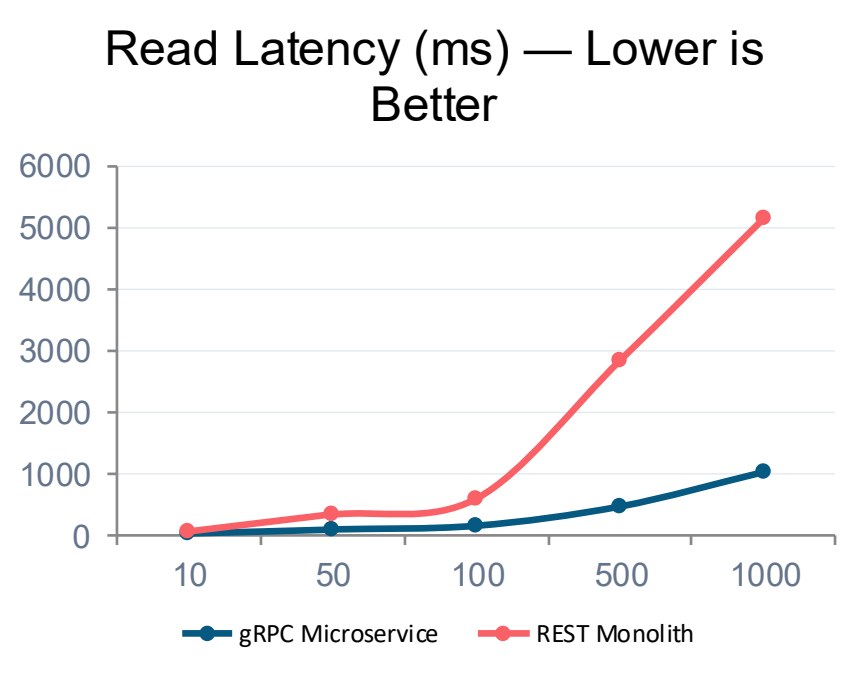
Metrics: Avg Latency (ms) + Throughput (req/s)

```
Latency = T_response - T_request  Throughput = Total Requests / Total elapsed Time
```

*gRPC: 6 nodes | REST Monolith: 2 containers | Both running simultaneously*

# Performance Results




**At 1000 users: gRPC read latency 1023ms vs REST 5150ms — 5× better | gRPC write throughput 63% higher**

# Trade-off Analysis

| Dimension | gRPC Microservice | REST Monolith |
|---|---|---|
| Performance | Higher throughput, lower latency | Degrades sharply at scale |
| Scalability | Horizontal via load balancer | Single bottleneck |
| Fault Tolerance | Redundant servers + DB replication | Single point of failure |
| Complexity | Higher — multi-node orchestration | Lower — one container |
| Dev Speed | Slower — proto definitions needed | Faster — standard REST |

# AI Tools — How Claude Helped

### Architecture Design

Designed the 6-node system layout and Docker Compose configuration

### Proto File

Generated all gRPC .proto definitions for 6 service methods

### Debugging

Diagnosed Dockerfile COPY path error and fixed build context

### Benchmarking

Generated concurrent Python benchmark and plotting scripts

### Documentation

Wrote README, final report, and this presentation

*Prompting Strategy: provide exact errors + architecture constraints → targeted, working solutions*

# Conclusion

- ✅ 6 functional requirements fully implemented and tested
- ✅ 6-node gRPC microservice architecture deployed with Docker
- ✅ gRPC delivers 5× lower latency and 63% higher throughput at scale
- ✅ Microservice design provides fault tolerance REST cannot match
- ✅ AI tools (Claude) accelerated development at every stage

**Thank you! | Q&A**